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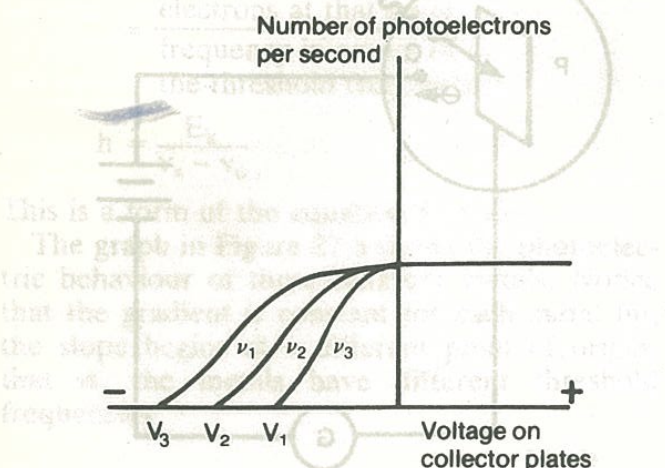


Figure 27.13

These graphs show that the current flow in a photoelectric cell is proportional to the intensity of the light falling on the metallic plate. They also show that altering the frequency of the incident light will affect the cut-off voltage of the cell.

Different metals will have different characteristics when used in photoelectric cells. Experiments with different metals produce data similar to that shown in Figure 27.14. As previously stated, the gradient of each of these lines would be the constant h , Planck's constant.

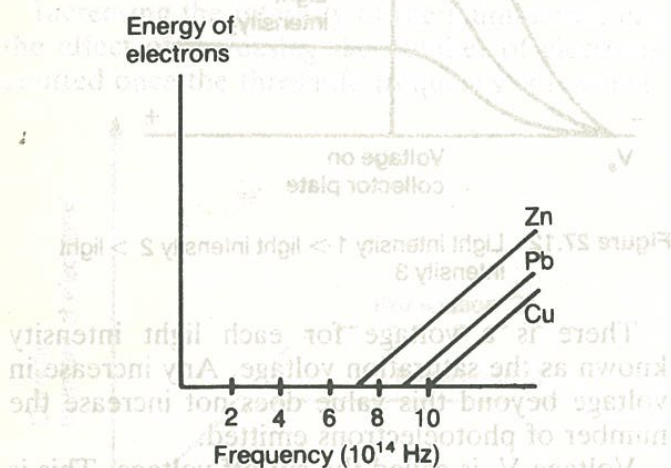


Figure 27.14 Photoelectric characteristics of different metals

Review exercise

- 1 Explain the concept of the work function of a metal. Why do different metals have different work functions?
- 2 What is meant by the cut-off voltage of a photoelectric cell?

3 The alkali metal rubidium has a threshold frequency of 5.0×10^{14} Hz.

- a What is the work function for rubidium?
- b What is the maximum velocity with which photoelectrons will be ejected when the rubidium surface is subjected to radiation of wavelength 3.15×10^{-7} m?

Major ideas

- Rutherford's planetary model of the atom failed to explain observable physical phenomena and so had to be modified.
- Bohr modified Rutherford's model of the atom by suggesting that
 - 1 Electrons can only exist in certain energy levels.
 - 2 As long as an electron remains in its lowest energy level, no energy is emitted.
 - 3 Energy is only emitted when an electron jumps from a higher energy level to a lower energy level. This energy is given by the expression

$$E_2 - E_1 = h\nu = \Delta E.$$

- When an excited electron returns to the ground state it can only do so by certain allowable energy jumps. These jumps will be different for atoms of different elements and lead to characteristic line spectra for each element.
- The emission of electrons by photoelectric metals when they are subjected to light is governed by
 - 1 The frequency required for a photon to eject an electron from a metal surface is called the threshold frequency. Each photon can only eject one electron.
 - 2 A minimum amount of work must be done to remove an electron from a metal. The value of this work is called the work function.
 - 3 The energy of a photoelectron is controlled by the frequency of the photon, and unaffected by the intensity of the light.
 - 4 The number of electrons ejected depends on the intensity of the light but is not affected by the frequency.
 - 5 The energy of the photon not used to free the electron from the metal gives the electron its kinetic energy.
 - 6 The energy of the ejected photoelectrons

is given by the expression

$$E = h\nu = \phi + E_{k \text{ max}}$$

Revision questions

1 The electron in the hydrogen atom can exist at various energy levels, the three lowest being $0.70 \times 10^{-19} \text{ J}$, $17.0 \times 10^{-19} \text{ J}$ and $20.0 \times 10^{-19} \text{ J}$, with respect to the ground level $0.70 \times 10^{-19} \text{ J}$.

a How many lines can be predicted in the spectrum of atomic hydrogen? What are their photon energies?

b Which of these lines has the shortest wavelength? Calculate the wavelength.

2 Light of wavelength $4.50 \times 10^{-7} \text{ m}$ falls on a clean metal surface which has a work function of $5.7 \times 10^{-17} \text{ J}$. Does photoelectric emission occur? Show how you arrived at your answer.

3 How does Bohr's model of the atom explain the emission of energy by atoms?

4 Explain why the spectrum of hydrogen can contain so many lines when the hydrogen atoms has only one electron.

5 One ultraviolet photon will eject an electron from zinc, but a hundred photons of blue light will not. Explain this.

6 How can an atom be raised to an excited state?

7 In the photoelectric effect of metals, from where does the energy to eject the photoelectrons come? Into what forms is the energy given to the photoelectrons divided?

8 If the frequency of incident light is less than the threshold frequency for the photoelectric metal, where does the photon energy go?

9 Can photoelectrons be distinguished from electrons from another source? Explain your answer.

10 What is meant by the term photoelectron? Explain a practical use of the photoelectric effect.

In order to be electrically neutral, an atom must contain the same number of protons, that is positive charges, as orbital electrons, that is negative charges. Experiments carried out by Ernest Rutherford lead to the conclusion that the positive charges must be concentrated in the nucleus of the atom.

The positively charged protons in the nucleus remain together in the nucleus in spite of the repulsive electric forces that act between them. This is possible because a nuclear force that is stronger than the electric force acts between all particles in the nucleus. The nuclear force is complicated and a precise mathematical description of it has not been obtained. It is a very strong force over a short distance of about 10^{-12} m . At distances greater than this it is practically zero.

A stable nucleus is one whose nuclear forces are such that it stays together indefinitely.

It was found that the nuclear charge of atoms of a particular element is always the same and is unique for that element. The nuclear charge on the hydrogen nucleus being the smallest for any element, is called the unit nuclear charge. The name given to the nuclear particle carrying this unit charge is the proton. The number of protons in a nucleus, and therefore the amount of positive charge, is called the atomic number.

The positive charge on a proton has the same magnitude as the charge on an electron. For the proton it is $+1.6 \times 10^{-19} \text{ C}$, while for the electron it is $-1.6 \times 10^{-19} \text{ C}$. This means that the atomic number also gives the number of non-nuclear electrons in a neutral atom. The atomic number is represented by the letter Z , and it is

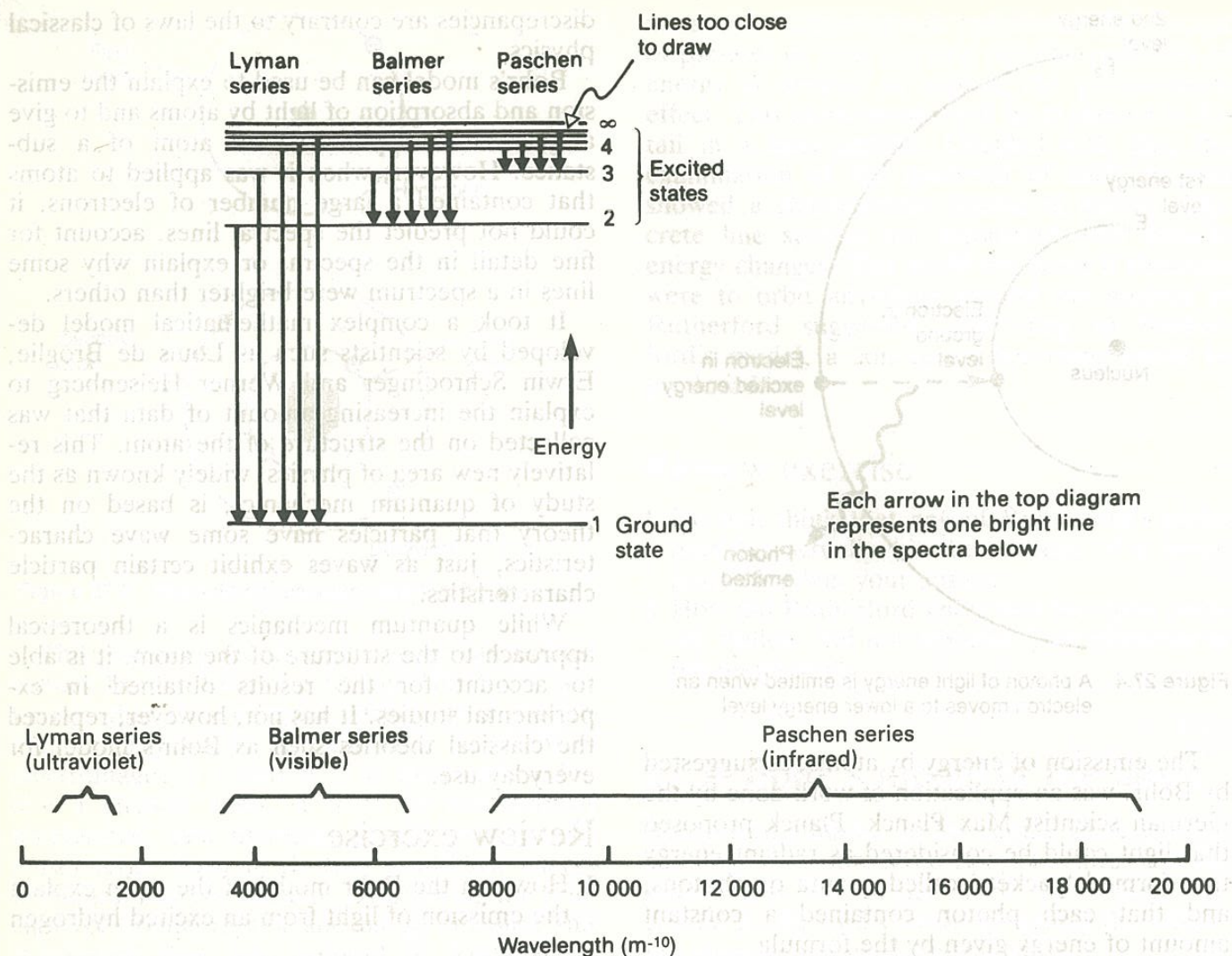


Figure 27.5 The production of line spectra in hydrogen.

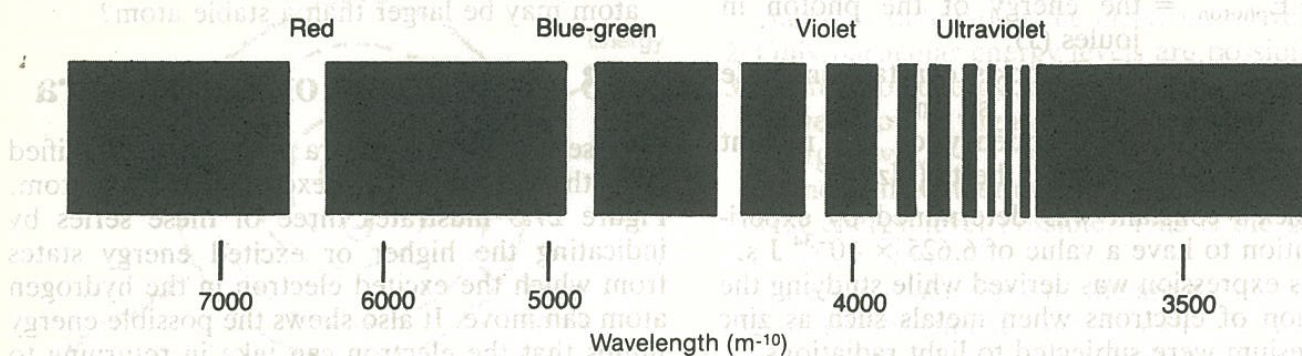


Figure 27.6 The Balmer series in hydrogen

Energy is only emitted at the moment the electron jumps from a higher energy level to a lower energy level. The amount of energy emitted can be determined using the following formula:

$$\text{Energy emitted} = h\nu$$

$$h\nu = E_2 - E_1$$

where h = Planck's constant in joule seconds (J s), $6.625 \times 10^{-34} \text{ J s}$,
 ν = the frequency of the radiation in hertz (Hz), and
 E_2 and E_1 = the energy of the higher energy level and lower energy level in joules respectively (J).

Example

What is the frequency and wavelength of the light emitted when an electron falls from an energy level of 7.92×10^{-19} J to a level of 4.23×10^{-19} J of energy?

To calculate the frequency

$$h = 6.625 \times 10^{-34} \text{ J s}$$

$$E_2 = 7.92 \times 10^{-19} \text{ J}$$

$$E_1 = 4.23 \times 10^{-19} \text{ J}$$

$$\nu = ?$$

$$h\nu = E_2 - E_1$$

$$6.625 \times 10^{-34} \times \nu = 7.92 \times 10^{-19} - 4.23 \times 10^{-19}$$

$$\nu = \frac{7.92 \times 10^{-19} - 4.23 \times 10^{-19}}{6.625 \times 10^{-34}}$$

$$= \frac{3.69 \times 10^{-19}}{6.625 \times 10^{-34}}$$

$$= 5.57 \times 10^{14} \text{ Hz}$$

To calculate the wavelength of the light emitted

$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$

$$\nu = 5.57 \times 10^{14} \text{ Hz}$$

$$\lambda = ?$$

$$c = \lambda \nu$$

$$3.00 \times 10^8 = \lambda \times 5.57 \times 10^{14}$$

$$\lambda = \frac{3.00 \times 10^8}{5.57 \times 10^{14}}$$

$$= 5.38 \times 10^{-7} \text{ m}$$

The frequency and wavelength of the emitted light are 5.57×10^{14} Hz and 5.38×10^{-7} m respectively.

Review exercise

- 1 Use a diagram to explain how the Bohr model accounts for the visible spectrum of hydrogen gas.
- 2 If an atom has five energy levels, how many spectral lines could it emit? Which of the transitions would produce the photon of highest energy?
- 3 The wavelength of a red line which is visible in the Balmer series of the hydrogen spectrum is 6.56×10^{-7} m. What is the difference in energy between the two energy levels responsible for this spectral line?

27.4 The photoelectric effect

When light falls on metals such as zinc and cesium, electrons can be ejected from the surface

of the metal. This ejection of electrons is called the photoelectric effect. The energy absorbed from the incident radiation enables the electrons to increase their kinetic energy sufficiently to escape the attractive forces of the nuclei in the surface of the metal.

For each metal that exhibits the photoelectric property there is specific minimum energy (E_0) required to release electrons from the metal. Hence there is a minimum frequency (ν_0), known as the threshold frequency, which will release the electrons. These quantities are related by the expression

$$E_0 = h\nu_0$$

where E_0 = the minimum energy in joules (J),

h = Planck's constant, 6.625×10^{-34} joule second (J s), and

ν_0 = the threshold frequency in hertz (Hz).

It has been found that even the weakest illumination will release electrons if the frequency is equal to, or greater than, the threshold frequency for the metal. If the frequency is not great enough, then even the most intense illumination does not have any effect. The value of the threshold frequency depends upon the individual metal. Increasing the frequency beyond the threshold frequency increases the maximum kinetic energy of the ejected electron. This behaviour can be illustrated graphically as shown in Figure 27.7. No electrons are emitted until the threshold frequency is reached. At frequencies beyond this, the kinetic energy of the electrons increases. The gradient of this graph is equal to the constant h , Planck's constant.

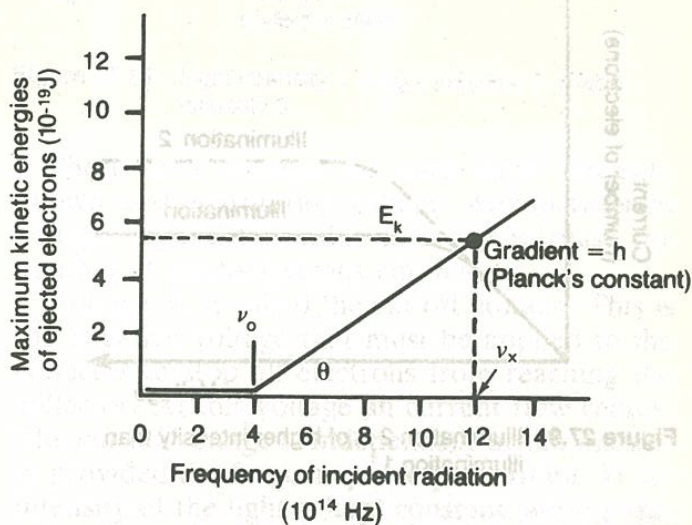


Figure 27.7 Kinetic energy of ejected electrons versus frequency of incident radiation

gradient = $\tan \theta$

kinetic energy of the
electrons at that point
= frequency in excess of
the threshold frequency

$$h = \frac{E_k}{\nu_x - \nu_0}$$

This is a form of the equation $E = h\nu$.

The graph in Figure 27.8 shows the photoelectric behaviour of three different metals. Notice that the gradient is constant for each metal but the slope begins at a different point of origin, that is, the metals have different threshold frequencies.

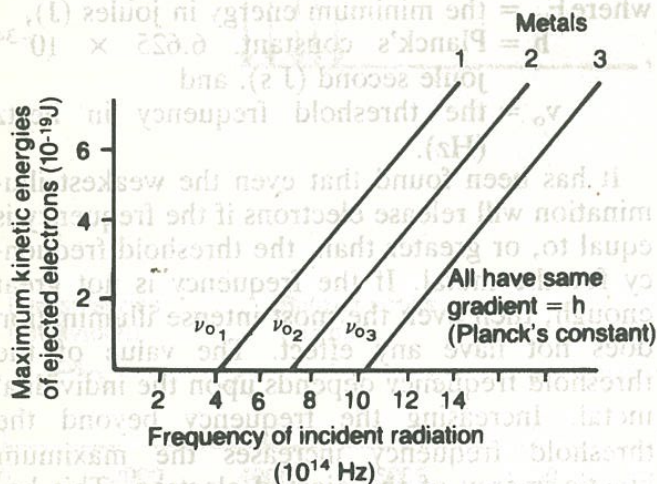


Figure 27.8 Photoelectric characteristics for different metals

Increasing the intensity of the illumination has the effect of increasing the number of electrons emitted once the threshold frequency is reached.

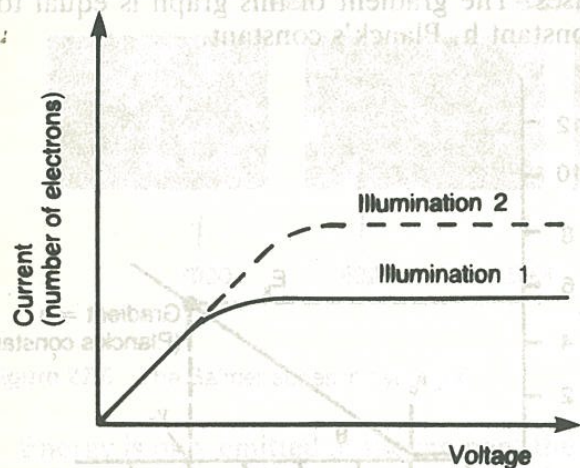


Figure 27.9 Illumination 2 is of higher intensity than illumination 1

This increase in electrons emitted can be detected by measuring the change in current flow

from the metal to a positively charged plate as the intensity of illumination is altered. There is a maximum amount of current that will flow for each value. Figure 27.9 shows how the current flow is altered as the voltage on the positively charged plate is varied.

Einstein explained the function of the threshold frequency by suggesting that for each photoelectric metal there is a certain amount of work which must be done to remove an electron from the surface of the material. This minimum quantity of work is called the work function, and varies for different metals. If an electron is to be emitted, the metal must receive energy equivalent to the threshold energy and this energy must be received in one burst or quanta, not as an accumulation of smaller amounts. If more energy is received than the threshold energy, the balance goes towards increasing the kinetic energy of the photoelectron. In this way, energy is conserved. This idea can be put in the form of an equation

Energy of incident photon = energy required to remove electron from metal + maximum kinetic energy of ejected electron.

$$E = h\nu = \text{work function} + \text{maximum kinetic energy}$$

$$E = h\nu = \phi + E_{k \text{ max}}$$

where E = the energy in joules (J),

h = Planck's constant, 6.625×10^{-34} joule second (J s),

ν = the frequency in hertz (Hz),

ϕ = the work function in joules (J), and

$E_{k \text{ max}}$ = the maximum kinetic energy in joules (J).

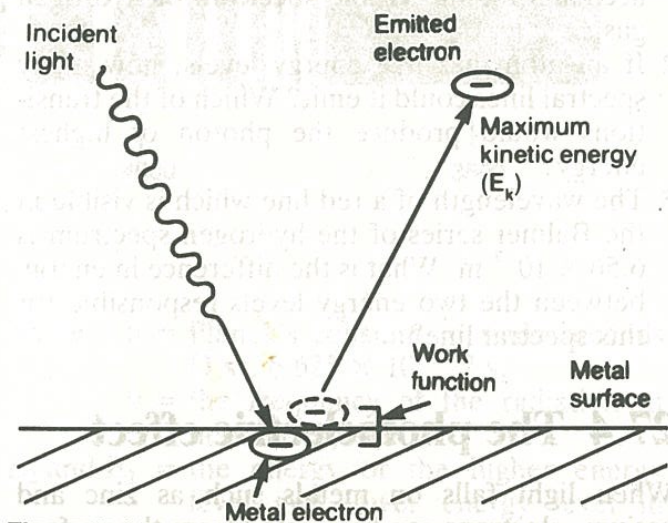


Figure 27.10

For alkali metals the work function is small, hence relatively low frequency electromagnetic radiation such as visible light ($10^{14} - 10^{15}$ Hz) can emit photoelectrons. Most other metals require a higher frequency electromagnetic radiation such as ultraviolet radiation ($10^{15} - 10^{16}$ Hz).

Example

Electromagnetic radiation of wavelength 1.5×10^{-7} m illuminates a metal surface which has a work function of 6.1×10^{-19} J. What is the energy of the emitted photoelectrons?

Determine the frequency of the incident radiation.

$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$

$$\lambda = 1.5 \times 10^{-7} \text{ m}$$

$$\nu = ?$$

$$c = \lambda \nu$$

$$3.00 \times 10^8 = 1.5 \times 10^{-7} \times \nu$$

$$\nu = 2.0 \times 10^{15} \text{ Hz}$$

Determine the energy of the emitted photoelectrons.

$$\nu = 2.0 \times 10^{15} \text{ Hz}$$

$$h = 6.625 \times 10^{-34} \text{ J s}$$

$$\phi = 6.1 \times 10^{-19} \text{ J}$$

$$E_k = ?$$

$$h\nu = \phi + E_{k \text{ max}}$$

$$6.625 \times 10^{-34} \times 2.00 \times 10^{15} = 6.1 \times 10^{-19} + E_k$$

$$13.2 \times 10^{-19} = 6.1 \times 10^{-19} + E_k$$

$$E_{k \text{ max}} = (13.2 - 6.1) \times 10^{-19}$$

$$= 7.1 \times 10^{-19} \text{ J}$$

Maximum energy of emitted photoelectrons is 7.1×10^{-19} J.

The device that is used to investigate the photoelectric effect is called a photoelectric cell. It consists of an electron-emitting metal plate and an electron collector enclosed in an evacuated transparent glass bulb. The collector is connected to the positive terminal of a power supply and a galvanometer is used to detect current flow as shown in Figure 27.11.

When light strikes the metallic plate P of the photoelectric cell, electrons are emitted and these are attracted to the positively charged collector C. This movement of electrons is detected as a flow of current by the galvanometer G. No current will flow if the photoelectric cell is covered so that no light falls on the plate.

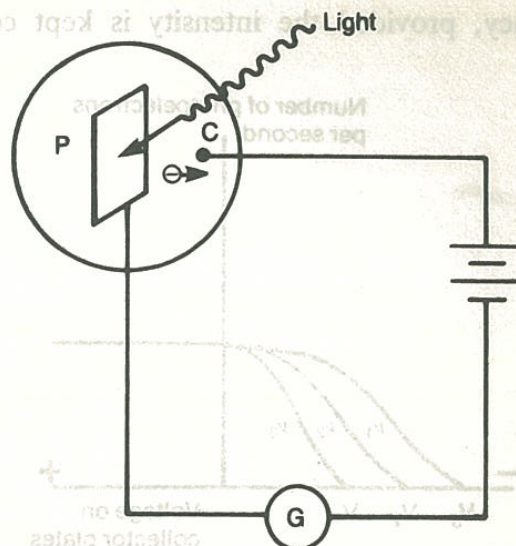


Figure 27.11 Vacuum photoelectric cell

The effect of making the collector of the photoelectric cell more positive can be illustrated by Figure 27.12. The intensity of the light was altered each time but the colour and frequency was kept constant.

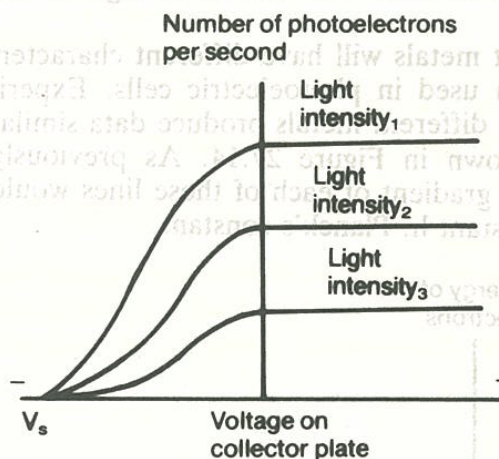


Figure 27.12 Light intensity 1 > light intensity 2 > light intensity 3

There is a voltage for each light intensity known as the saturation voltage. Any increase in voltage beyond this value does not increase the number of photoelectrons emitted.

Voltage V_s is called the cut-off voltage. This is the negative voltage that must be applied to the collector to stop all electrons from reaching the collector. At this voltage all current flow ceases. The cut-off voltage is independent of the intensity provided the frequency is kept constant. If the intensity of the light is kept constant and the frequency is altered, the following graph is produced. The cut-off voltage is dependent on the

frequency, provided the intensity is kept constant.

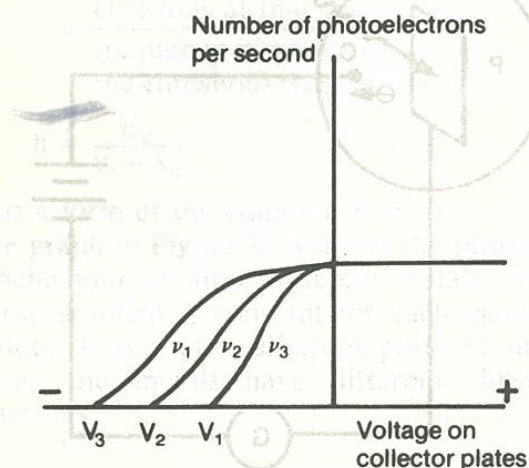


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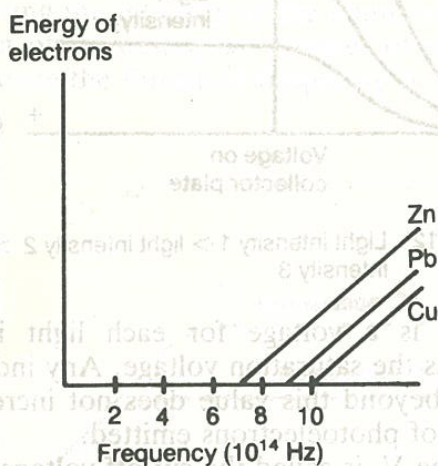


Figure 27.14 Photoelectric characteristics of different metals

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3 The alkali metal rubidium has a threshold frequency of 5.0×10^{14} Hz.

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